



Cambridge (CIE) IGCSE Biology



Your notes

Cell Structure & Size of Specimens

Contents

- * Cell Structure
- * Organisation of Cells
- * Magnification Formula
- * Converting Between Units



Your notes

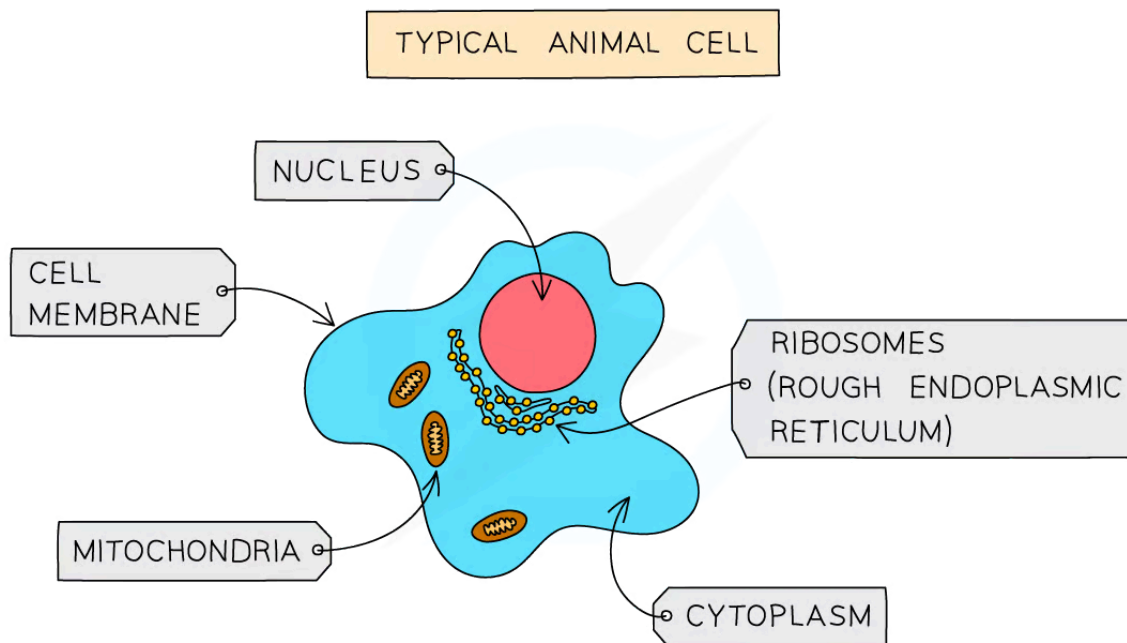
Cell Structure

Animal & plant cells

Animal cell structure

- The main features of animal cells:
 - They contain a **nucleus** with a **distinct membrane**
 - Cells **do not** have **cellulose cell walls**
 - Their cells **do not** contain **chloroplasts** (so they **are unable** to carry out **photosynthesis**)
 - They contain carbohydrates stored as **glycogen**

Animal cell diagram



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A typical animal cell

Plant cell structure

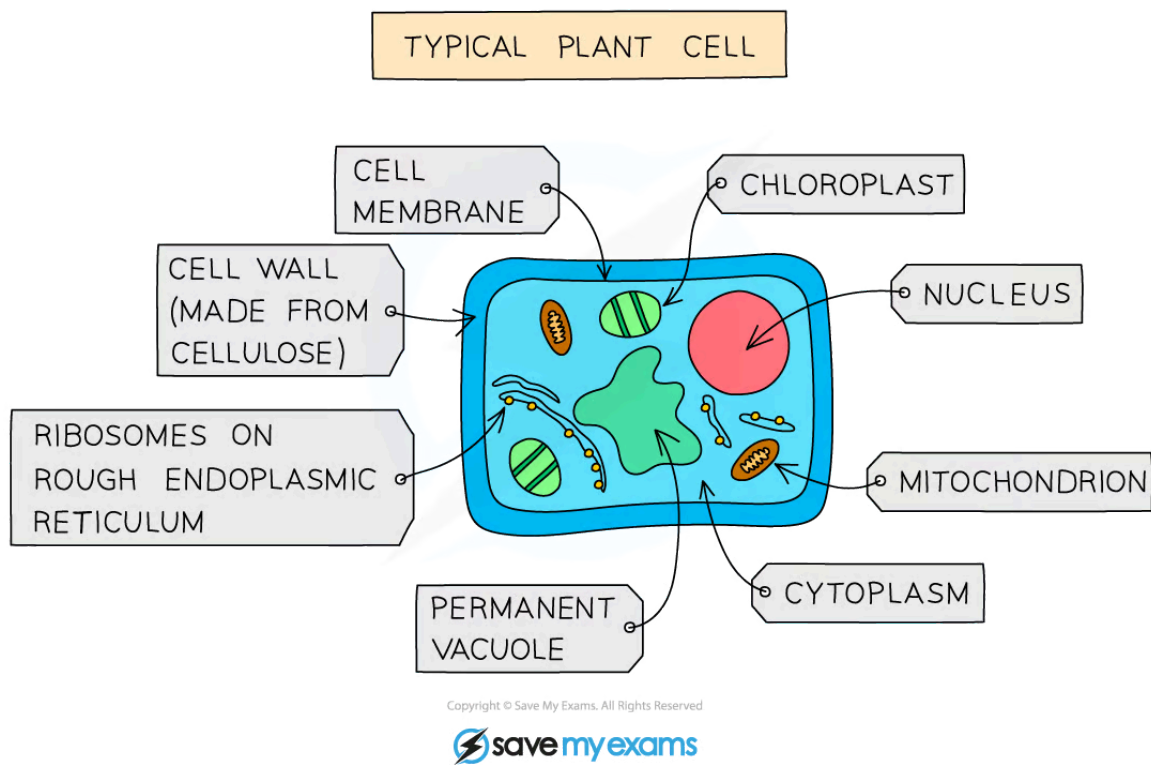


Your notes

▪ The main features of plant cells:

- They contain a **nucleus** with a **distinct membrane**
- Cells have **cell walls** made out of **cellulose**
- They contain **chloroplasts** (so they can carry out **photosynthesis**)
- Carbohydrates are stored as **starch** or **sucrose**

Plant cell diagram



A typical plant cell

Plant and animal cell structure and function

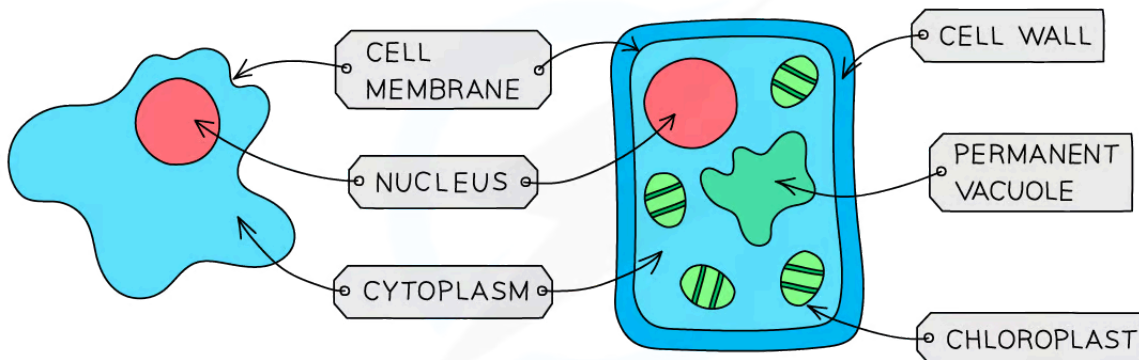
Structure	Function
Nucleus	▪ Contains the DNA (genetic material) which controls the activities of the cell



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Cytoplasm	- A gel like substance composed of water and dissolved solutes - Supports the internal structures of the cell - Site of many chemical reactions (including anaerobic respiration)
Cell membrane	- Holds the cell together separating the inside of the cell from the outside - Controls which substances enter or leave the cell
Ribosomes	- Found in the cytoplasm - The site of protein synthesis
Mitochondria	The site of aerobic respiration

Cell structure diagram



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An animal and plant cell as seen under a light microscope

Plant cell structure and function

Structure	Function
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Your notes

Cell wall	- Made of cellulose (a polymer of glucose) - Gives the cell extra support, defining its shape
Chloroplast	Contains the green chlorophyll pigment that absorbs light energy for photosynthesis
Permanent vacuole	- Contains cell sap: a solution of sugar and salt - Used for storage of certain materials - Helps to support the shape of the cell

Bacteria cells

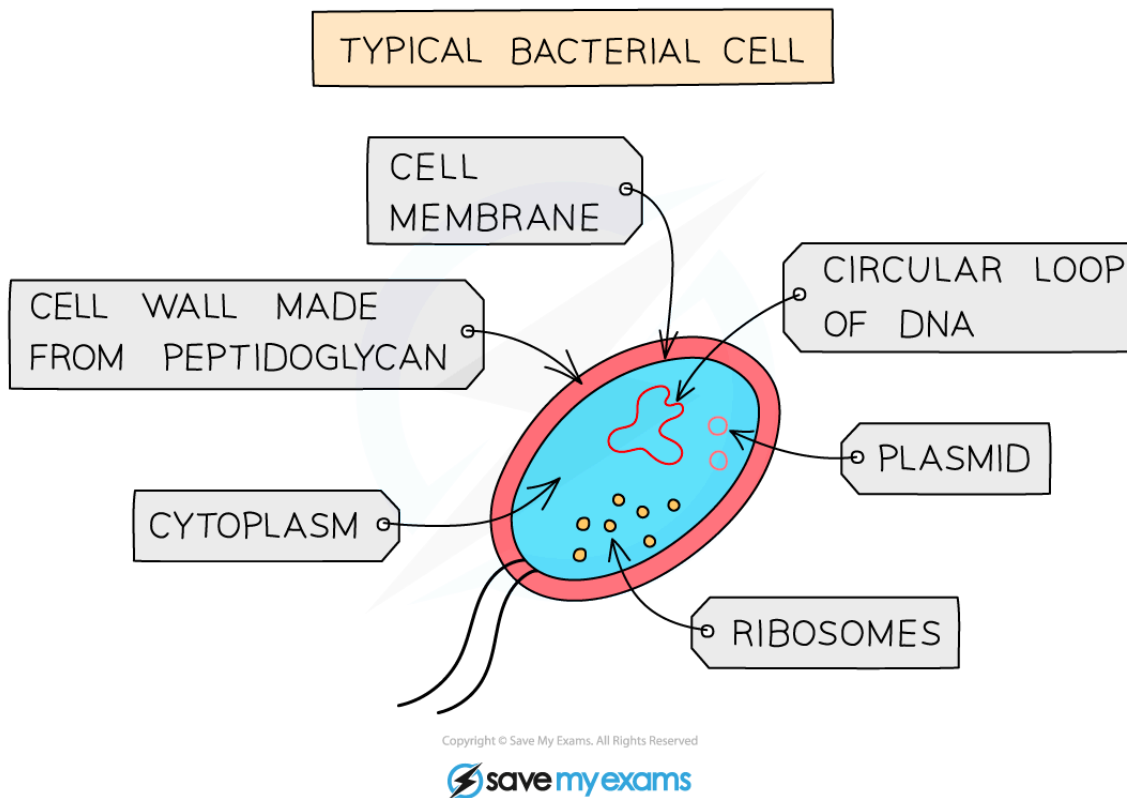
Bacteria cell structure

- Bacteria, which have a wide variety of shapes and sizes, all share the following biological characteristics:
 - They are **microscopic single-celled organisms**
 - Possess a **cell wall** (made of **peptidoglycan**, not cellulose), **cell membrane**, **cytoplasm** and **ribosomes**
 - Lack a nucleus** but contain a **circular chromosome of DNA that floats in the cytoplasm**
 - Plasmids** are sometimes present - these are **small rings of DNA** (also floating in the cytoplasm) that contain **extra genes** to those found in the chromosomal DNA
 - They **lack mitochondria, chloroplasts and other membrane-bound organelles** found in animal and plant cells
- Some bacteria also have a **flagellum** (singular) or **several flagella** (plural). These are **long, thin, whip-like tails** attached to bacteria that allow them to **move**
- Examples of bacteria include:
 - Lactobacillus** (a rod-shaped bacterium used in the production of yoghurt from milk)
 - Pneumococcus** (a spherical bacterium that acts as the pathogen causing pneumonia)

Bacteria cell diagram



Your notes



A typical bacterial cell

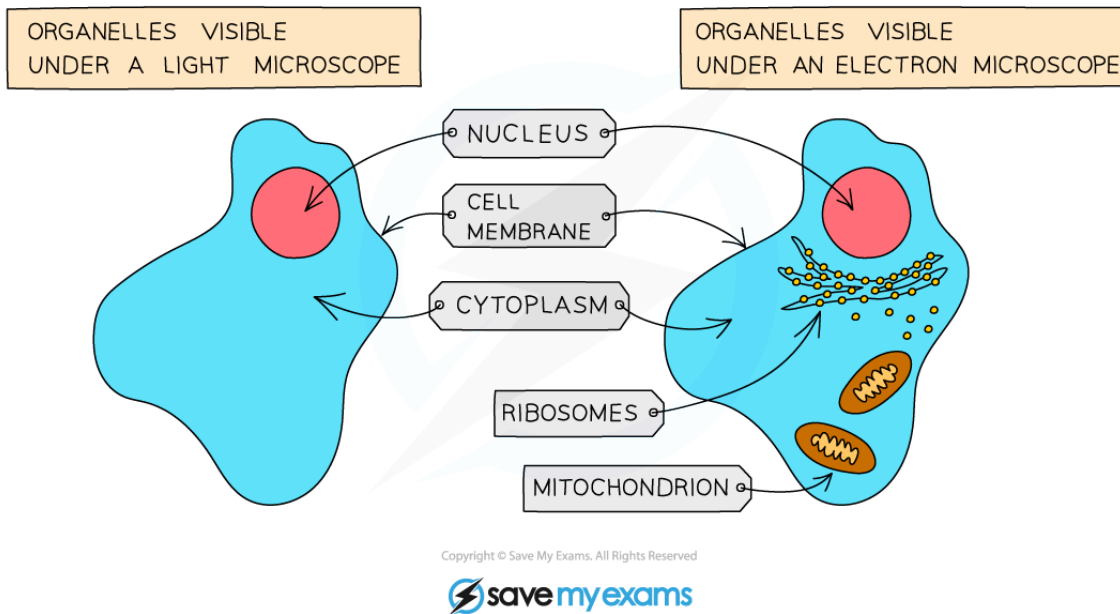
Identifying cell structures & function

- Within the cytoplasm, the following organelles are visible in almost all cells except prokaryotes when looking at higher magnification (ie using an electron microscope):
 - **Mitochondria** (singular: mitochondrion) are organelles found throughout the cytoplasm
 - **Ribosomes** are tiny structures that can be free within the cytoplasm or attached to a system of membranes within the cell known as **Endoplasmic Reticulum**
 - Endoplasmic reticulum studded with ribosomes looks **rough** under the microscope; this gives rise to its name of **Rough Endoplasmic Reticulum** (often shortened to **R.E.R.**)
 - **Vesicles** can also be seen using a higher magnification - these are small circular structures found moving throughout the cytoplasm

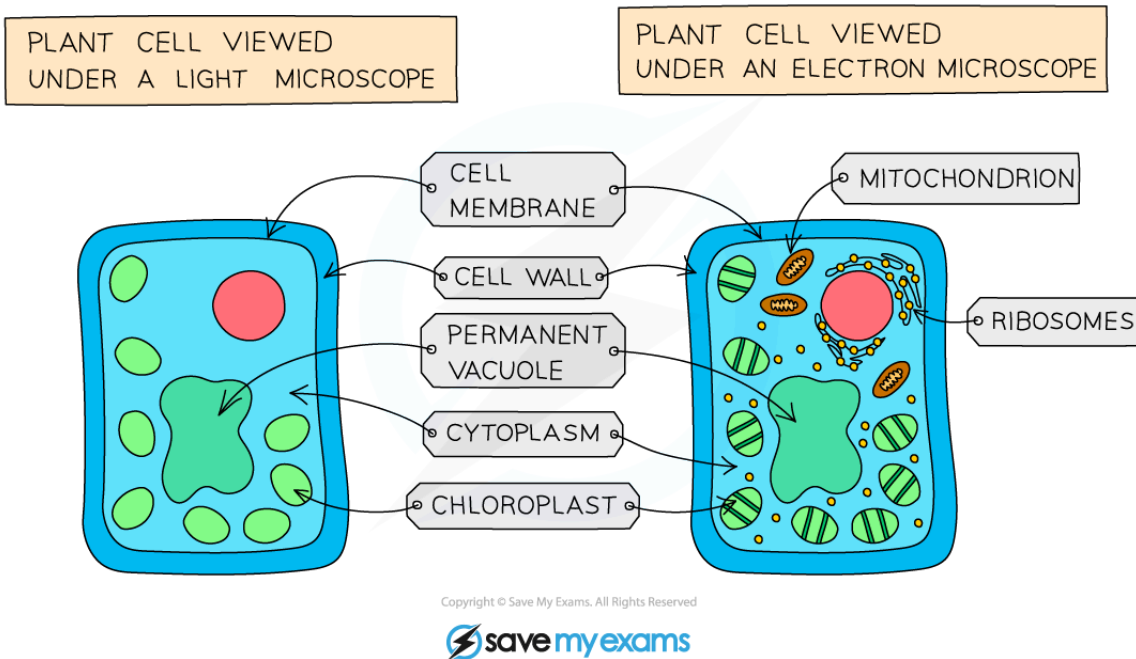
Identifying cell structures under a microscope



Your notes



Structures in an animal cell visible under a light microscope and an electron microscope



Structures in a plant cell visible under a light microscope and an electron microscope



Your notes



Your notes

Organisation of Cells

Producing New Cells

- The cells in your body need to be able to divide to help your body **grow** and **repair** itself
- Cells grow and divide over and over again
- New cells are produced by the **division of existing cells**

Specialised Cells

Specialised cells in animals

- Specialised cells are those which have **developed certain characteristics** in order to **perform particular functions**. These differences are controlled by genes in the nucleus
- Cells specialise by undergoing **differentiation**: this is a process by which cells develop the structure and characteristics needed to be able to carry out their functions

Specialised Cells in Animals Table



Your notes

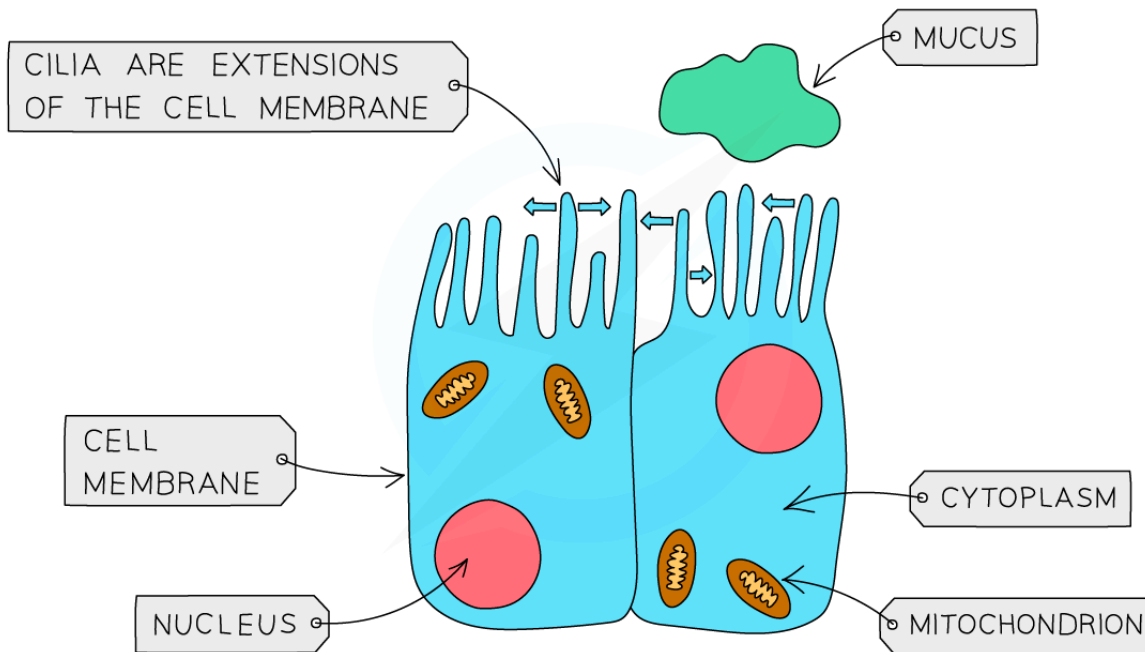
CELL	FUNCTION	ADAPTATIONS
CILIATED CELL	MOVEMENT OF MUCUS IN THE TRACHEA AND BRONCHI	EXTENSIONS OF THE CYTOPLASM AT THE SURFACE OF THE CELL FORM HAIR-LIKE STRUCTURES CALLED CILIA WHICH BEAT TO MOVE MUCUS AND TRAPPED PARTICLES UP TO THE THROAT
NERVE CELL	CONDUCTION OF IMPULSES	LONG SO THAT NERVES CAN RUN TO AND FROM DIFFERENT PARTS OF THE BODY TO THE CENTRAL NERVOUS SYSTEM - THE CELL HAS EXTENSIONS AND BRANCHES, SO THAT IT CAN COMMUNICATE WITH OTHER NERVE CELLS, MUSCLES AND GLANDS - THE AXON (EXTENSION OF CYTOPLASM AWAY FROM THE CELL BODY) IS COVERED WITH A FATTY SHEATH, WHICH INSULATES THE NERVE CELL AND SPEEDS UP THE NERVE IMPULSE
RED BLOOD CELL	TRANSPORT OF OXYGEN	BICONCAVE DISC SHAPE INCREASES SURFACE AREA FOR MORE EFFICIENT DIFFUSION OF OXYGEN - CONTAINS HAEMOGLOBIN WHICH JOINS WITH OXYGEN TO TRANSPORT IT - CONTAINS NO NUCLEUS TO INCREASE AMOUNT OF SPACE AVAILABLE FOR HAEMOGLOBIN INSIDE CELL
SPERM CELL	REPRODUCTION	- THE HEAD CONTAINS THE GENETIC MATERIAL FOR FERTILISATION IN A HAPLOID NUCLEUS (CONTAINING HALF THE NORMAL NUMBER OF CHROMOSOMES) - THE ACROSOME IN THE HEAD CONTAINS DIGESTIVE ENZYMES SO THAT A SPERM CAN PENETRATE AN EGG - THE MID-PIECE IS PACKED WITH MITOCHONDRIA TO RELEASE ENERGY NEEDED TO SWIM AND FERTILISE THE EGG - THE TAIL ENABLES THE SPERM TO SWIM
EGG CELL (OVUM)	REPRODUCTION	- CONTAINS A LOT OF CYTOPLASM WHICH HAS NUTRIENTS FOR THE GROWTH OF THE EARLY EMBRYO HAPLOID NUCLEUS CONTAINS THE GENETIC MATERIAL FOR FERTILISATION - CELL MEMBRANE CHANGES AFTER FERTILISATION BY A SINGLE SPERM SO THAT NO MORE SPERM CAN

ENTER



Your notes

Diagrams of specialised cells in animals:



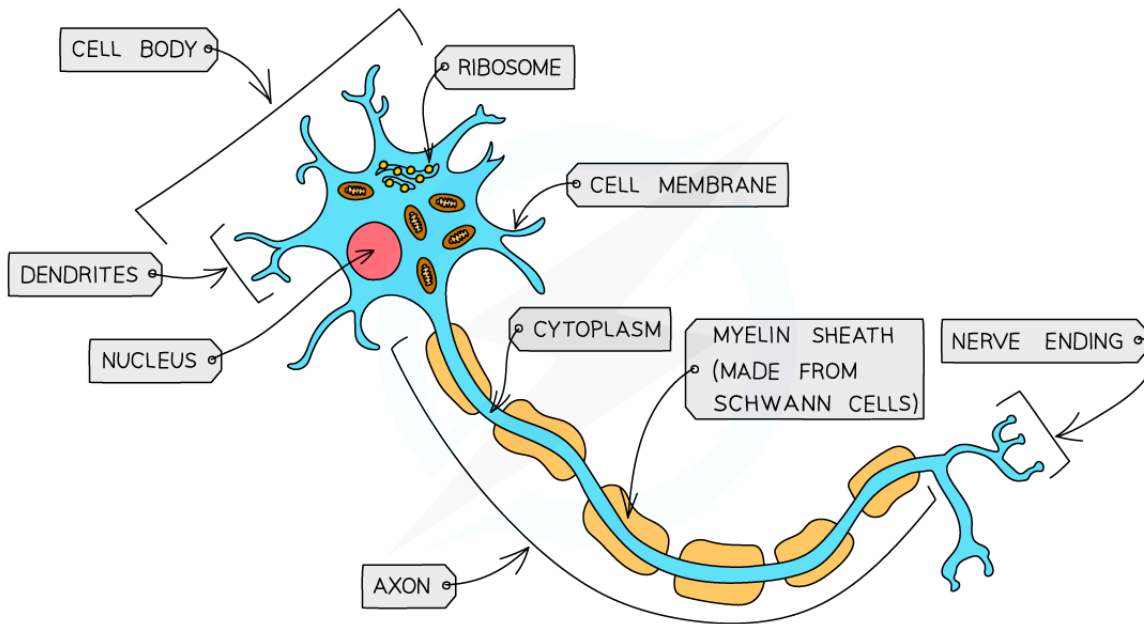
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Ciliated cell



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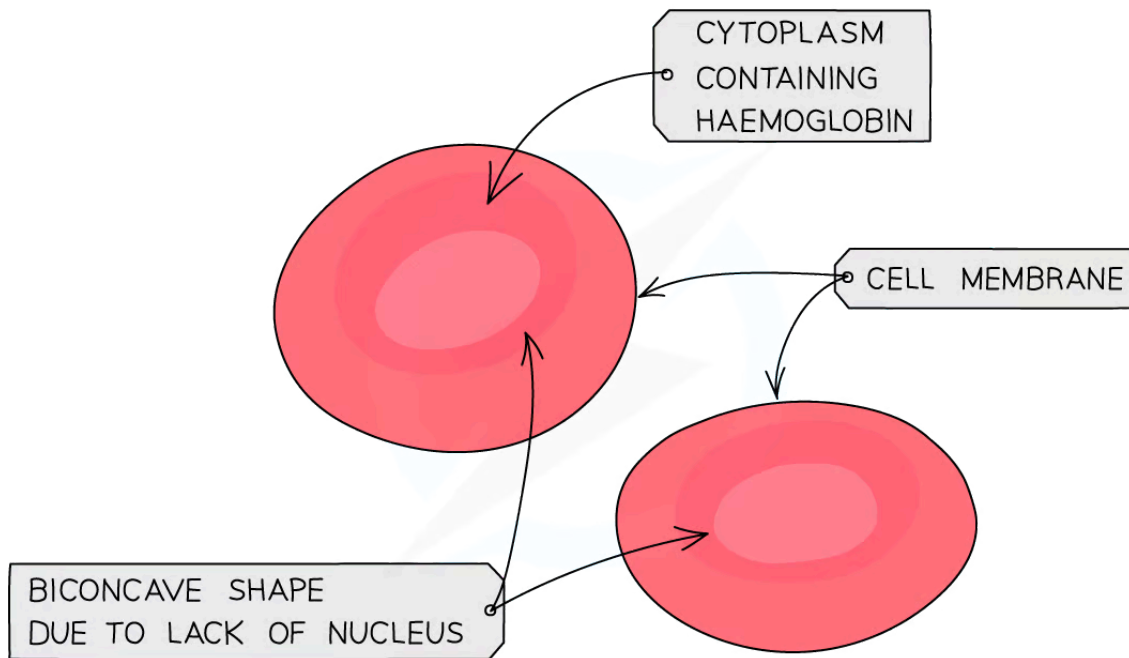
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Nerve cell



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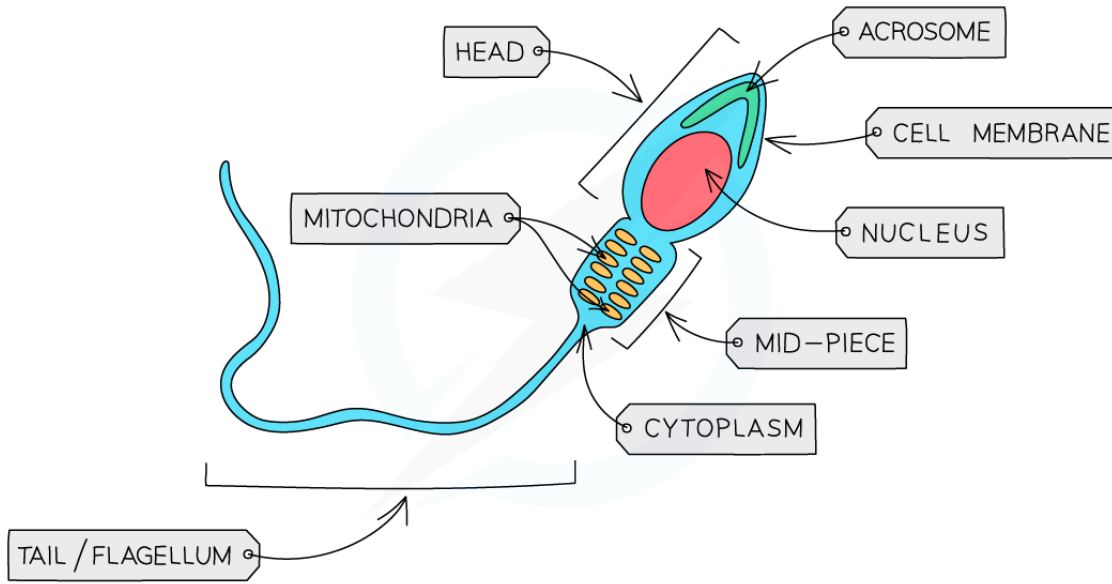
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Red blood cells



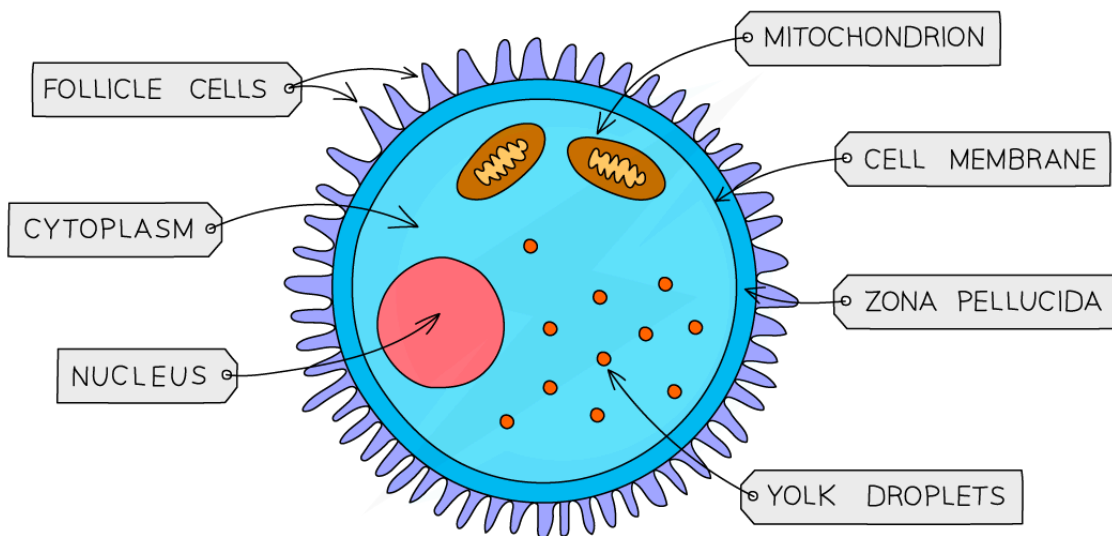
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Sperm cell



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Egg cell

Examples of specialised cells in plants:



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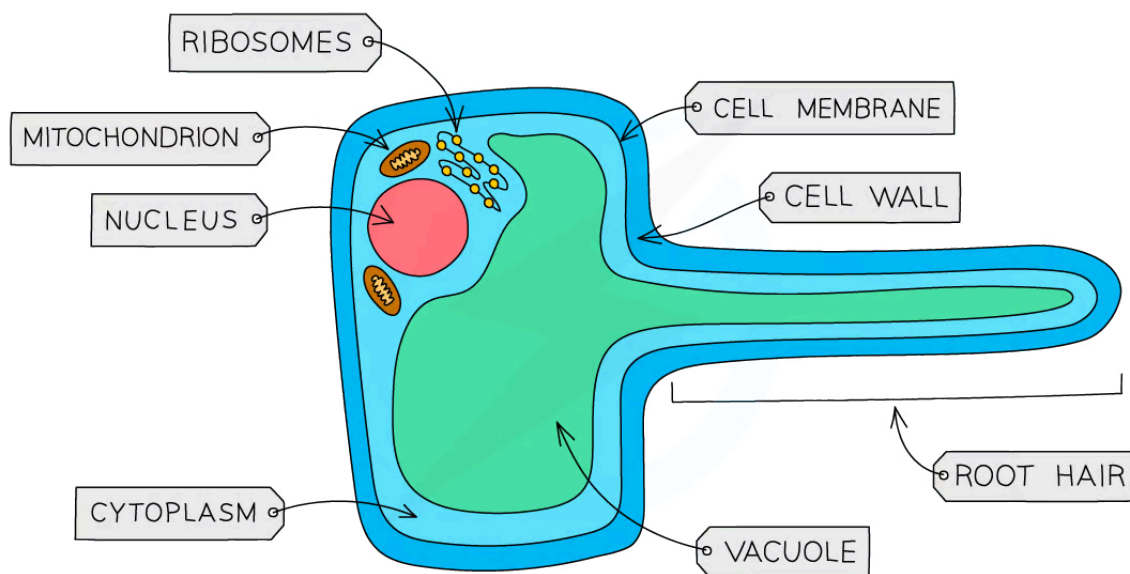
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ROOT HAIR CELL	ABSORPTION OF WATER AND MINERAL IONS FROM SOIL	– ROOT HAIR INCREASES SURFACE AREA OF CELL TO ENSURE MAXIMUM ABSORPTION OF WATER AND MINERAL IONS – WALLS ARE THIN TO ENSURE WATER MOVES THROUGH QUICKLY – NO CHLOROPLASTS PRESENT
XYLEM VESSEL	CONDUCTION OF WATER THROUGH THE PLANT; SUPPORT OF THE PLANT	– NO TOP AND BOTTOM WALLS BETWEEN XYLEM VESSELS, SO THERE IS A CONTINUOUS COLUMN OF WATER RUNNING THROUGH THEM – CELLS ARE DEAD WITHOUT ORGANELLES OR CYTOPLASM TO ALLOW FREE PASSAGE OF WATER – THEIR WALLS BECOME THICKENED WITH A SUBSTANCE CALLED LIGNIN WHICH MEANS THEY ARE ABLE TO HELP SUPPORT THE PLANT
PALISADE MESOPHYLL CELL	PHOTOSYNTHESIS	– COLUMN SHAPED TO MAXIMIZE ABSORPTION OF SUNLIGHT AND FIT AS MANY IN A LAYER UNDER THE UPPER EPIDERMIS OF THE LEAF AS POSSIBLE – CONTAINS MANY CHLOROPLASTS FOR MAXIMUM PHOTOSYNTHESIS

Diagrams of specialised cells in plants:



Your notes



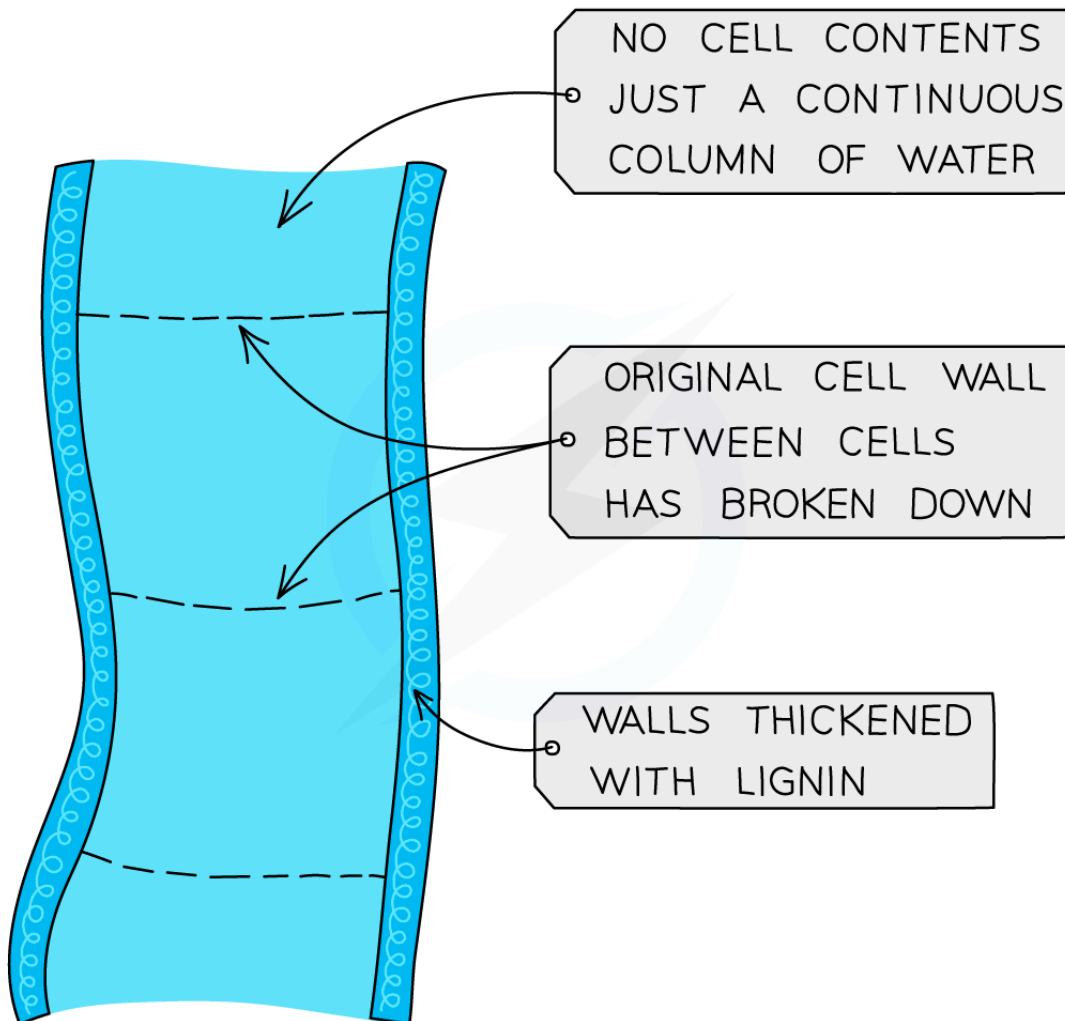
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Root hair cell



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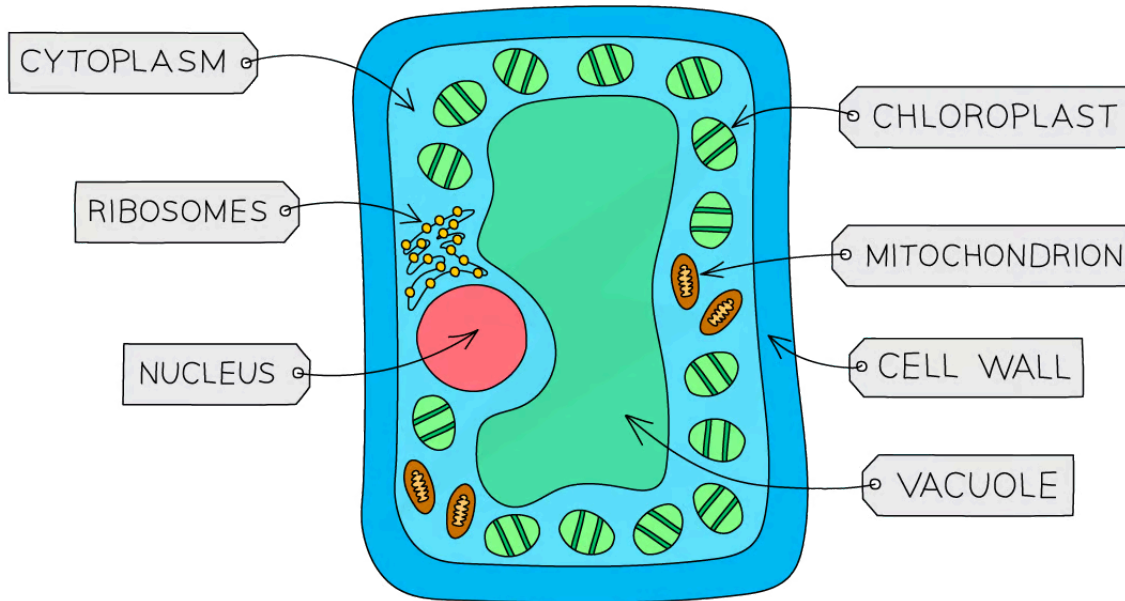
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Xylem structure



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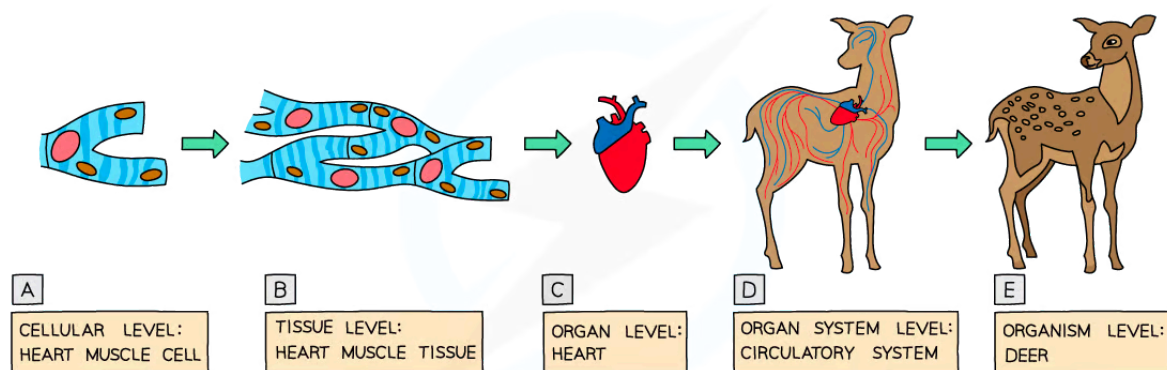
Palisade mesophyll cell

Levels of Organisation in an Organism



Your notes

LEVEL	DESCRIPTION
CELLS	BASIC FUNCTIONAL AND STRUCTURAL UNITS IN A LIVING ORGANISM
TISSUES	GROUPS OF CELLS OF SIMILAR STRUCTURE WORKING TOGETHER TO PERFORM THE SAME FUNCTION
ORGANS	MADE FROM DIFFERENT TISSUES WORKING TOGETHER TO PERFORM SPECIFIC FUNCTIONS
ORGAN SYSTEMS	GROUPS OF ORGANS WITH RELATED FUNCTIONS, WORKING TOGETHER TO PERFORM BODY FUNCTIONS



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Levels of organisation



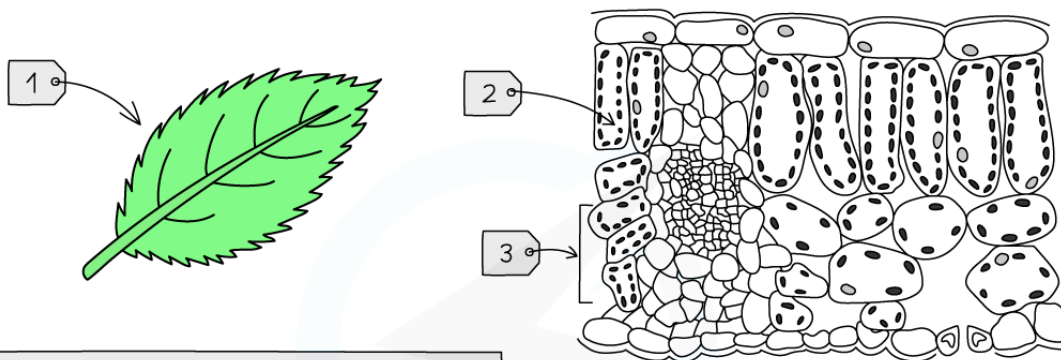
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ORGAN SYSTEM	ORGANS	TISSUE EXAMPLES
SHOOT SYSTEM	LEAF, STEM, FLOWER, FRUIT	– EPIDERMIS MESOPHYLL – XYLEM – PHLOEM
ROOT SYSTEM	ROOT, TUBER	– XYLEM – PHLOEM – GROUND TISSUE
DIGESTIVE SYSTEM	OESOPHAGUS, STOMACH, SMALL INTESTINE, LARGE INTESTINE	– MUSCLE – CONNECTIVE – NERVE – EPITHELIAL
CIRCULATORY SYSTEM	HEART, VEINS, ARTERIES	– MUSCLE – CONNECTIVE – NERVE – EPITHELIAL
IMMUNE SYSTEM	THYMUS, SPLEEN	– BONE MARROW
RESPIRATORY SYSTEM	TRACHEA, BRONCHI, LUNGS	– CONNECTIVE – MUSCLE – EPITHELIAL
EXCRETORY SYSTEM	LIVER, KIDNEY, SKIN, LUNGS	– MUSCLE – CONNECTIVE – EPITHELIAL – NERVE
NERVOUS SYSTEM	BRAIN, SPINAL CORD	– NERVE

REPRODUCTIVE SYSTEM	OVARY, CERVIX, UTERUS, VAGINA, TESTES, PENIS	– MUSCLE – CONNECTIVE – NERVOUS – ERECTILE
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- Your syllabus states that you should be able to identify the different levels of organisation in drawings, diagrams and images of familiar material
- An example of this is shown in the exam question below:

THE DIAGRAMS SHOW A LEAF AND ITS INTERNAL STRUCTURE



WHAT ARE THE LEVELS OF ORGANISATION OF THE LABELLED STRUCTURES?

	1	2	3
A	CELL	TISSUE	ORGAN SYSTEM
B	ORGAN	CELL	TISSUE
C	ORGAN SYSTEM	TISSUE	CELL
D	TISSUE	CELL	ORGAN

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Examiner Tips and Tricks

Most incorrect answers here come from not being able to identify a tissue, so it's worth making sure you understand and remember that **tissues are always made up of only one type of cell**.



Your notes



Your notes

Magnification Formula

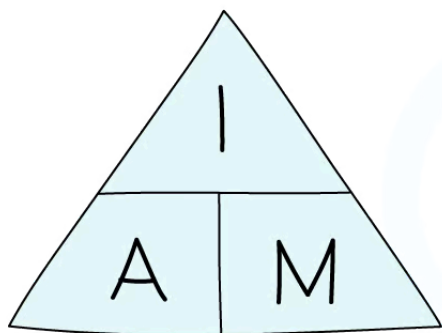
Magnification Formula

Calculating magnification and specimen size using millimetres as units

- Magnification is calculated using the following equation:

$$\text{Magnification} = \text{Image size} \div \text{Actual size}$$

- A better way to remember the equation is using an equation triangle:



WHERE: I = IMAGE / DRAWING SIZE
A = ACTUAL SIZE OF IMAGE
M = MAGNIFICATION

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Magnification equation

- Rearranging the equation to find things other than the magnification becomes easy when you remember the triangle - whatever you are trying to find, place your finger over it and whatever is left is what you do, so:
 - Magnification = image size / actual size
 - Actual size = image size / magnification
 - Image size = magnification x actual size

Remember magnification does not have any units and is just written as 'x 10' or 'x 5000'



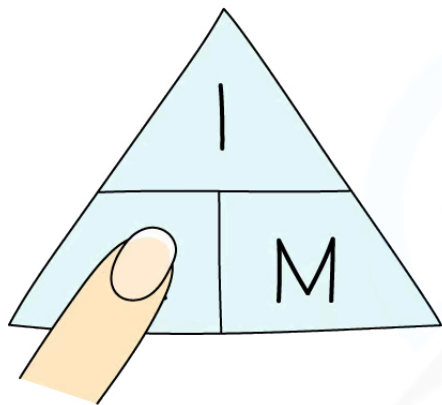


Your notes

Worked Example

An image of an animal cell is 30 mm in size and it has been magnified by a factor of $\times 3000$. What is the actual size of the cell?

To find the actual size of the cell:



$$A = \frac{I}{M} = \frac{30 \text{ mm}}{3000} = 0.01 \text{ mm}$$

$$0.01 \text{ mm} = 10 \mu\text{m}$$

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Worked example using the magnification equation



Examiner Tips and Tricks

This skill most frequently comes up in paper 5 and 6 (although it also comes up in the multiple choice and occasionally the theory paper) and you will definitely have to **calculate either magnification, drawing size or actual size** in at least one paper. To ensure you do not lose marks:

1. Always **look at the units** that have been given in the question - if you are asked to measure something, most often you will be expected to measure it in millimetres NOT in centimetres - double check the question to see!
2. **Learn the equation triangle for magnification** and write it on the page straight away
3. Don't forget that **magnification has NO UNITS** - students often lose a mark because they put one in



Your notes

Converting Between Units

Converting Between Units: Extended

Extended Tier Only

Using millimetres and micrometres as units

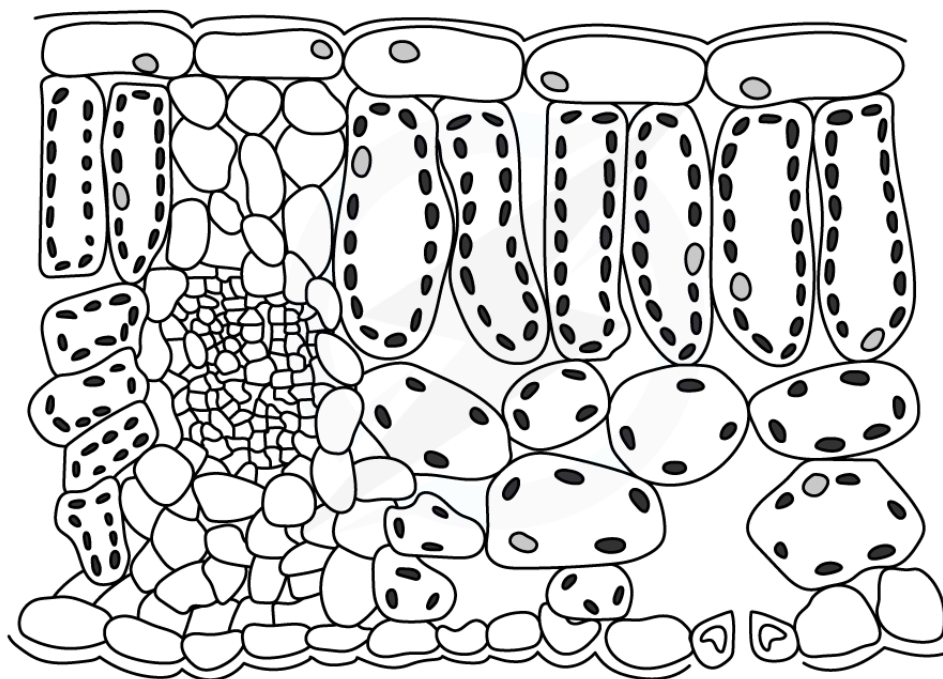
- The table below shows how millimetres are related to two other measures of length
- What this basically means is that $1\text{ mm} = 1000\mu\text{m}$ and $1\text{ cm} = 10,000\mu\text{m}$
- This usually comes up in questions where you have two different units and you need to ensure that you convert them both into the same unit before proceeding with the calculation
- For example:

UNIT	LENGTH IN mm
1 CENTIMETRE (cm)	10 mm
1 MILLIMETRE (mm)	1 mm
1 MICROMETRE (μm)	0.001 mm



Your notes

THE ACTUAL THICKNESS OF THE LEAF BELOW IS $2000\mu\text{m}$, BUT THE IMAGE SIZE OF THE LEAF IN THE DIAGRAM IS 50mm



WHAT IS THE MAGNIFICATION OF THE DIAGRAM?

A $\times 0.025$ **B $\times 25$** C $\times 100$ D $\times 100\,000$

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Example extended magnification question

- Remember $1\text{mm} = 1000\mu\text{m}$

- $2000 / 1000 = 2$ so the actual thickness of the leaf is 2mm and the drawing thickness is 50mm
- Magnification = image size / actual size = $50 / 2 = 25$
- So the magnification is x25 (NO UNITS)

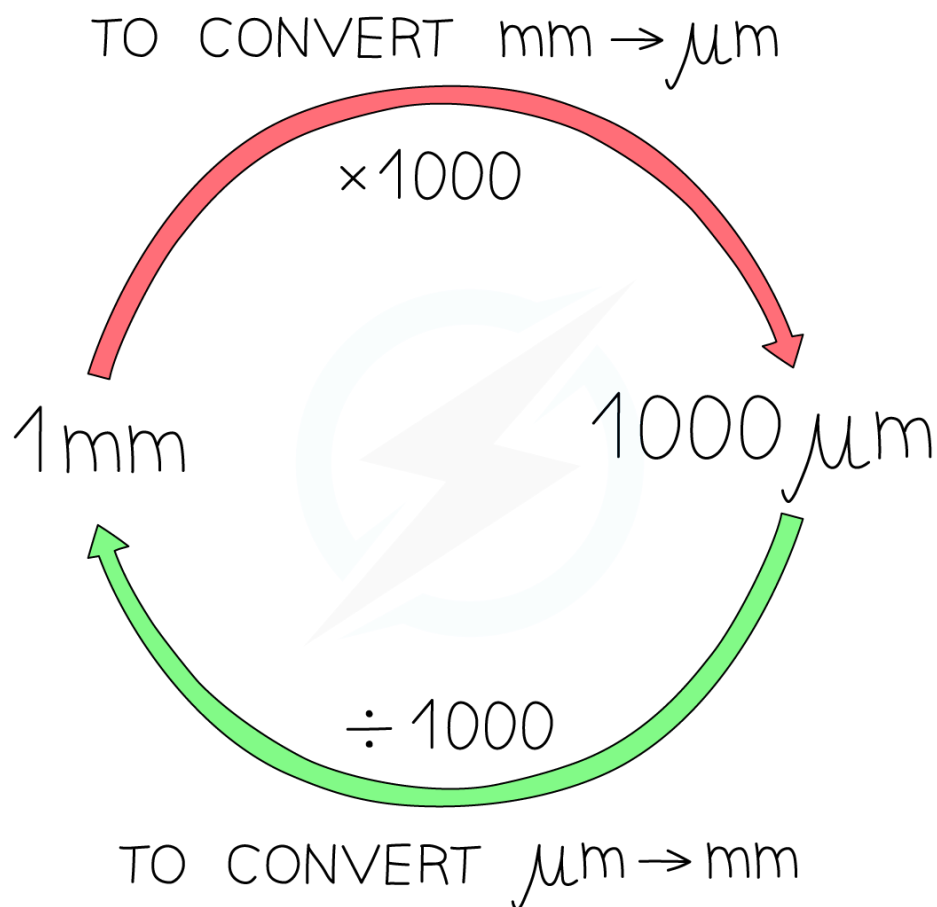


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Examiner Tips and Tricks

If you are given a question with two different units in it, make sure you convert them to the same unit before doing your calculation. If you don't, there is a good chance that your answer will be the same as one of the incorrect options in a multiple choice question so you may think you got it right when, in fact, you haven't! The following diagram may help with unit conversion between mm and μm :



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